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Jason Kountouridis

Research Interests

Arithmetic-algebraic geometry in positive and mixed characteristic

Singularities, monodromy and p-adic invariants related to birational geometry & number theory

Employment

2024-now **Simons postdoctoral fellow**, *Université Paris-Saclay & Institut de Mathématiques de Jussieu - Paris Rive Gauche*, Mentor: Kęstutis Česnavičius

Education

2018–2024 **PhD in Pure Mathematics**, *University of Chicago*, USA, Advisor: Matthew Emerton

2017–2018 **Master of Mathematics**, *University of Cambridge* (Part III), UK

2013–2017 **B.A. in Pure Mathematics**, *Princeton University*, USA

Publications

1. **Geometrically del Pezzo fibrations and canonical bundle formulas.** (with M. Benozzo) *In preparation - draft available.*
2. **Relative perversity of nearby cycles over general bases.** *Preprint. Draft available here.*
3. **Higher F-singularities for cones.** (with J. J. Garzella, V. Jagathese, R. Kurama and J. Witaszek) *Preprint. Draft available.*
4. **Monodromy of rational singularities on surfaces in mixed characteristic**. arXiv:2312.09175. *Newest version here. submitted.*
5. **Smoothness of components of the Emerton–Gee stack for GL_2.** (with A. Guzman, K. Kansal, B. Savoie and X. Wang) *Transactions of the American Mathematical Society, Vol. 378, pp. 67-115, 2025.*

Invited talks

Jun 2027 University of Athens, 11th Greek Conference on Algebra & Number Theory (upcoming)

Dec 2026 Université Sorbonne Paris Nord, Arithmetic Geometry Seminar (upcoming)

Sep 2026 Kaunas University of Technology, Conference dedicated to H. Minkowski

Sep 2026 Wuppertal, Summer School "Recent Developments in Arithmetic and Algebraic Geometry in Positive Characteristic"

Jun 2026 University of Athens, 3rd Congress of Greek Mathematicians TCGM 2026

Mar 2026 Institut de mathématiques de Jussieu, Number Theory Seminar

Mar 2026 Northwestern University, Algebraic Geometry Seminar

Jan 2025 CNRS, RéGA Seminar

Oct 2024 Université Paris-Saclay, SAGA Seminar (Séminaire Arithmétique et Géométrie Algébrique)

Feb 2024 University of Chicago, Number Theory Seminar

Feb 2024 Columbia University, Automorphic Forms and Arithmetic Seminar

Jan 2024 University of Wisconsin-Madison, Number Theory Seminar

Jan 2024 Northwestern University, Number Theory Seminar

Service

May 2027 **Université Paris-Saclay**, Co-organizer of the conference "Interactions between arithmetic and geometry in characteristic p" (upcoming)

2025-2026 **Université Paris-Saclay**, Co-organizer of a year-long seminar in birational geometry

2025 **Simons Collaboration**, Organizer of the postdoctoral learning seminar in the collaboration

2025 **Simons Collaboration**, Co-organizer of a reading group on Prismatic F-Gauges in the collaboration

2024-now **Université Paris-Saclay & Jussieu**, gave various talks in the Jussieu preprint seminar, a postdoctoral learning group in Paris-Saclay and other informal seminars

2020-2024 **University of Chicago**, organized the Graduate Number Theory Seminar in 2020-2021 along with various seminars on topics such as Arithmetic of K3 surfaces, Shimura Varieties and Deligne–Mumford Stacks

Conferences attended

Sep 2026 **Conference dedicated to H. Minkowski**, Kaunas University of Technology

Sep 2026 **Summer School "Recent Developments in Arithmetic and Algebraic Geometry in Positive Characteristic"**, Universität Wuppertal

Jul 2026 **Summer School on p-adic cohomology and explicit arithmetic**, Università di Genova

Jun 2026 **3rd Congress of Greek Mathematicians TCGM 2026**, University of Athens

Mar 2026 **Simons Collaboration Annual Meeting 2026**, New York

Oct 2025 **New structures and techniques in p-adic geometry**, IHES

Jul 2025 **2025 Summer Research Institute in Algebraic Geometry**, Colorado State University

Jun 2025 **Singularities in Plovdiv**, Plovdiv

Jun 2025 **p-adic and characteristic p methods in Algebraic Geometry**, EPFL, Lausanne

May 2025 **RGAS Summer School on Perfectoid Techniques**, La Cristalera

Mar 2025 **The Legacy of John Tate Memorial Conference**, Harvard University

Mar 2025 **Simons Collaboration Annual Meeting 2025**, New York

Oct 2024 **Methods in Mixed Characteristic Geometry**, Johannes Gutenberg University of Mainz

May 2024 **Singularities in Ann Arbor (Presented a poster)**, University of Michigan

Mar 2024 **Spring Workshop on p-adic Arithmetic Geometry**, Institute for Advanced Study, Princeton

Aug 2023 **Stacks Project Workshop**, University of Michigan

May 2023 **Algebraic Geometry and Cohomology in Mixed Characteristic** Northwestern University

Oct 2022 **Midwest Arithmetic Geometry and Number Theory Seminar**, University of Illinois, Chicago

Sep 2022 **Recent Perspectives in Hodge Theory**, Scuola Normale Superiore, Pisa

Mar 2022 **AWS 2022: Automorphic Forms Beyond $\mathbf{GL}_2$**, University of Arizona

Mar 2020 **AWS 2020: Nonabelian Chabauty**, University of Arizona

Oct 2019 **Midwest Arithmetic Geometry and Number Theory Seminar**, Ohio State University

Aug 2019 **Conference in Arithmetic and Algebraic Geometry**, University of Michigan

Awards & Scholarships

2017-2018 A.G. Leventis Foundation Fellowship

2017 Middleton Miller Prize for Best Independent Work in Mathematics, Princeton University

2016 Peter A. Greenberg '77 Prize for Outstanding Accomplishments in Mathematics, Princeton University

Teaching Experience

Spring 2027 **École Polytechnique**, Teaching Assistant for an L3 course in Differential Geometry (upcoming)

Spring 2026 **École Polytechnique**, Teaching Assistant for an L3 course in Differential Geometry

Fall 2025 **Université Paris–Saclay**, Supervised a project for multiple L2 Algebra students

Winter 2024 **University of Chicago**, **Instructor**: Math 15250 *Math Methods for Economic Analysis*

Fall 2023 **University of Chicago**, **Instructor**: Math 15250 *Math Methods for Economic Analysis*

Winter 2023 **University of Chicago**, **Instructor**: Math 15300 *Calculus III*

Fall 2022 **University of Chicago**, **Instructor**: Math 15250 *Math Methods for Economic Analysis*

Winter 2022 **University of Chicago**, **Instructor**: Math 15200 *Calculus II*

Fall 2021 **University of Chicago**, **Instructor**: Math 15100 *Calculus I*

2020-2021 **University of Chicago**, **Instructor**: Math 13000s *Elementary Functions and Calculus I-II-III*

2019-2020 **University of Chicago**, **College Fellow** (Teaching Assistant): Linear Algebra, Introduction to Proofs, Algebraic Number Theory

2015-2017 **Princeton University**, Grader and Teaching Assistant for undergraduate courses in Complex Analysis and Algebra

Languages

English (fluent), German (B2, intermediate), French (beginner), Greek (native)